

Robust IMF under contamination

Robust IMF changes the local fitting problem

Set $Y^{(1)} = Y$. At each stage k , extract the IMF component

$$\tilde{S}_t^{(k)} = \operatorname{arginf}_x \sum_u \rho(Y_u^{(k)} - x) K_{h_k}(t - u).$$

$K_h(v) = K(v/h)$ weights the local window. The contrast ρ penalizes $r = Y_u^{(k)} - x$; its derivative $\psi = \rho'$ determines each point's contribution to the fit.

Linear IMF

$$\rho_{\text{lin}}(r) = \frac{r^2}{2}, \quad \psi_{\text{lin}}(r) = r.$$

A weighted mean; large mismatches have large influence.

Subtract the component and shrink the bandwidth:

$$Y_t^{(k+1)} = Y_t^{(k)} - \tilde{S}_t^{(k)}, \quad h_{k+1} = h_k/a, \quad a = \sqrt{2}.$$

Robust IMF

$$\rho_H(r) = \mathbb{E}_\xi |r + H\xi|, \quad \xi \sim \mathcal{N}(0, 1),$$

$$\psi_H(r) = \operatorname{erf}\left(\frac{r}{\sqrt{2}H}\right), \quad |\psi_H(r)| \leq 1.$$

A smooth absolute-value contrast, fitted by gradient descent; $H = 2\sigma$.

Compare both methods on the same observations

Additive contamination

$$Y_i = X_i + G_i + M_i C_i.$$

Contaminated points retain the signal.

$G_i \sim \mathcal{N}(0, \sigma^2)$, $M_i \sim \text{Bernoulli}(p)$, and C_i is exponential noise with an independent random sign.

The RMSE curves and heatmaps compare each method m with its matching clean reference:

$$e_{k,m}(t_i) = \tilde{S}_m^{(k)}(Y)(t_i) - \tilde{S}_m^{(k)}(X)(t_i), \quad \text{RMSE}_{k,m} = \sqrt{\frac{1}{n} \sum_{i=1}^n e_{k,m}(t_i)^2}.$$

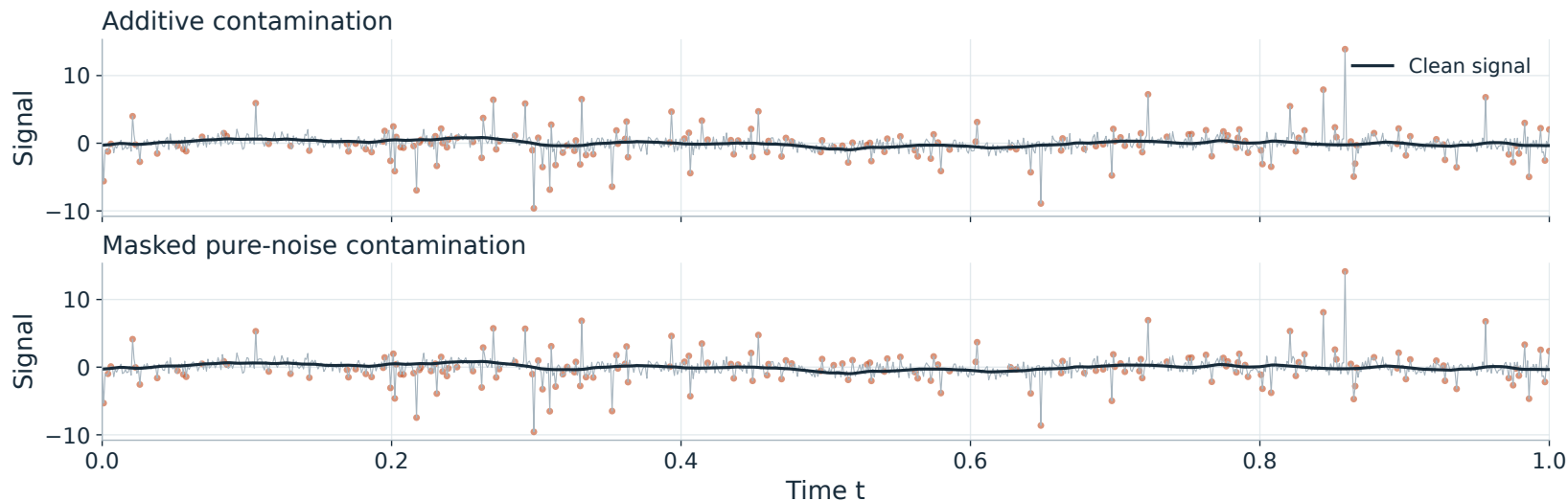
Both methods receive the same observation. Stage-comparison plots cover all nine stages.

Masked pure-noise contamination

$$Y_i = \begin{cases} X_i + G_i, & M_i = 0, \\ G_i + C_i, & M_i = 1. \end{cases}$$

Contaminated points lose the signal.

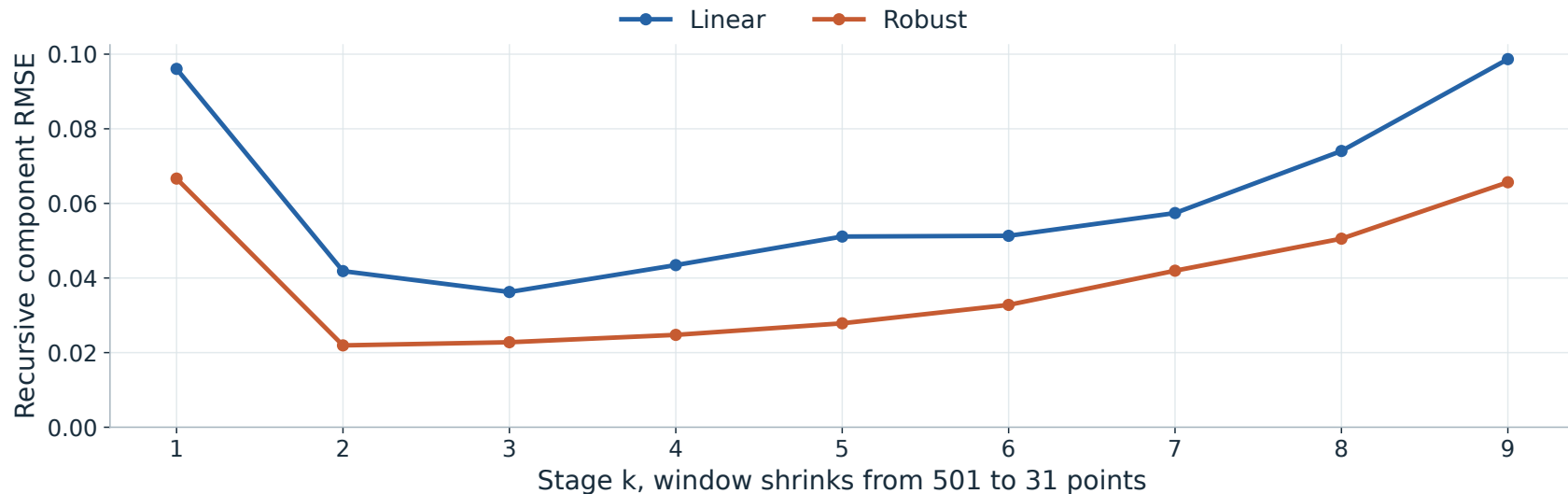
The contaminated observations



$n = 1000$, $\sigma = 0.6$, $p = 0.2$, contamination scale 2.0, robust $H = 1.2$.

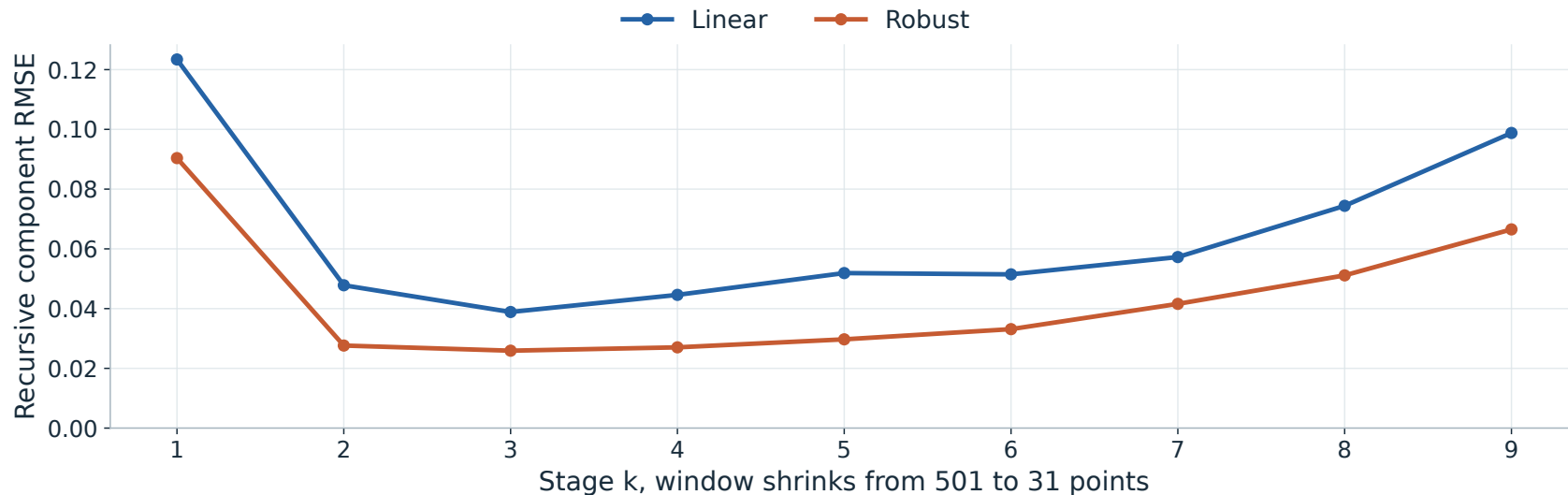
Nine windows shrink from 501 to 31 points. The 175 contaminated points are marked in orange.

Additive contamination at the default setting



Mean RMSE across the nine stages falls from 0.0611 to 0.0394, a 35.5% reduction.
Robust IMF has lower RMSE at all nine stages in this realization.

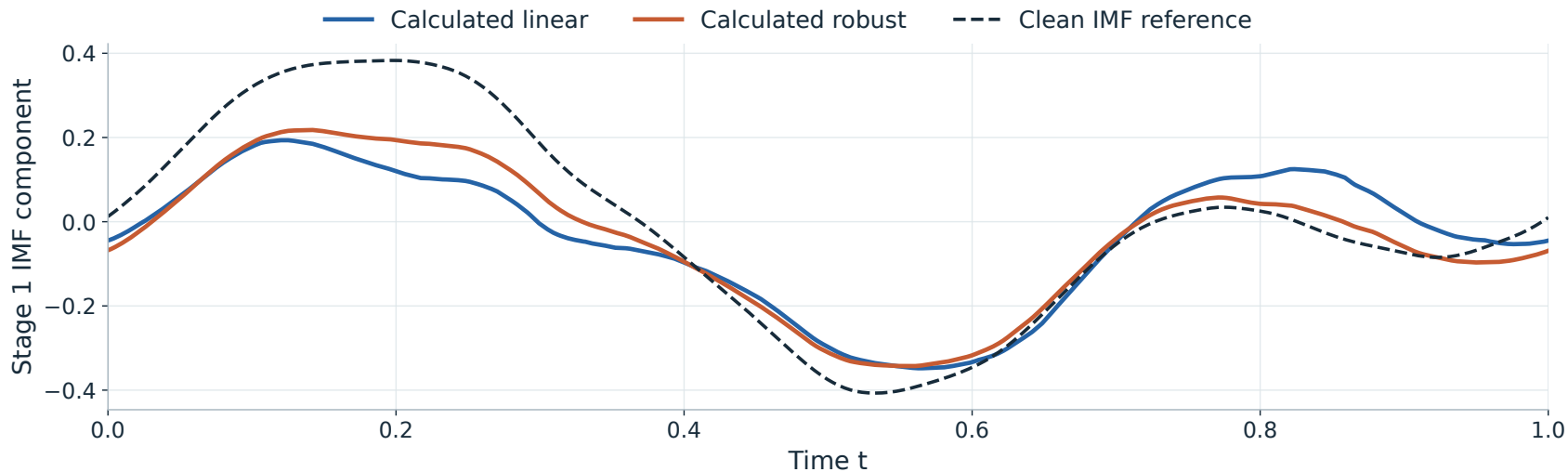
Masked contamination at the default setting



Mean RMSE falls from 0.0654 to 0.0437, a 33.2% reduction.

The robust advantage appears at every stage in this paired realization.

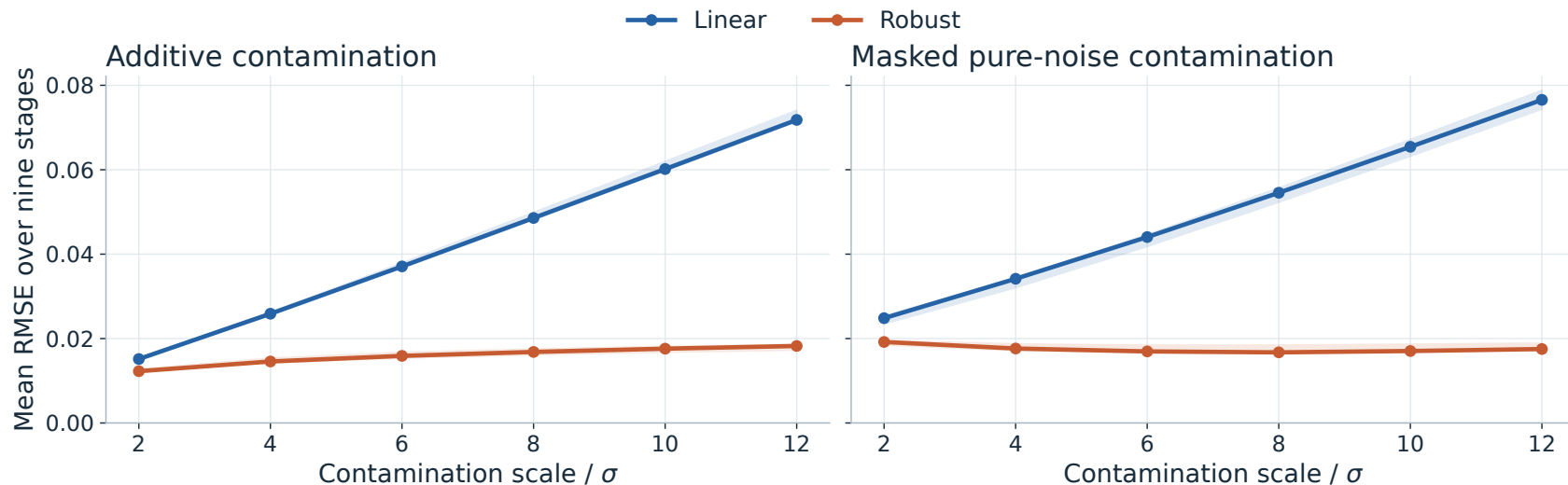
Robust IMF improves the first masked component



Both curves use the same clean linear IMF component as their reference.

Stage-1 RMSE is 0.1234 for linear IMF and 0.0898 for robust IMF, a 27.3% reduction.

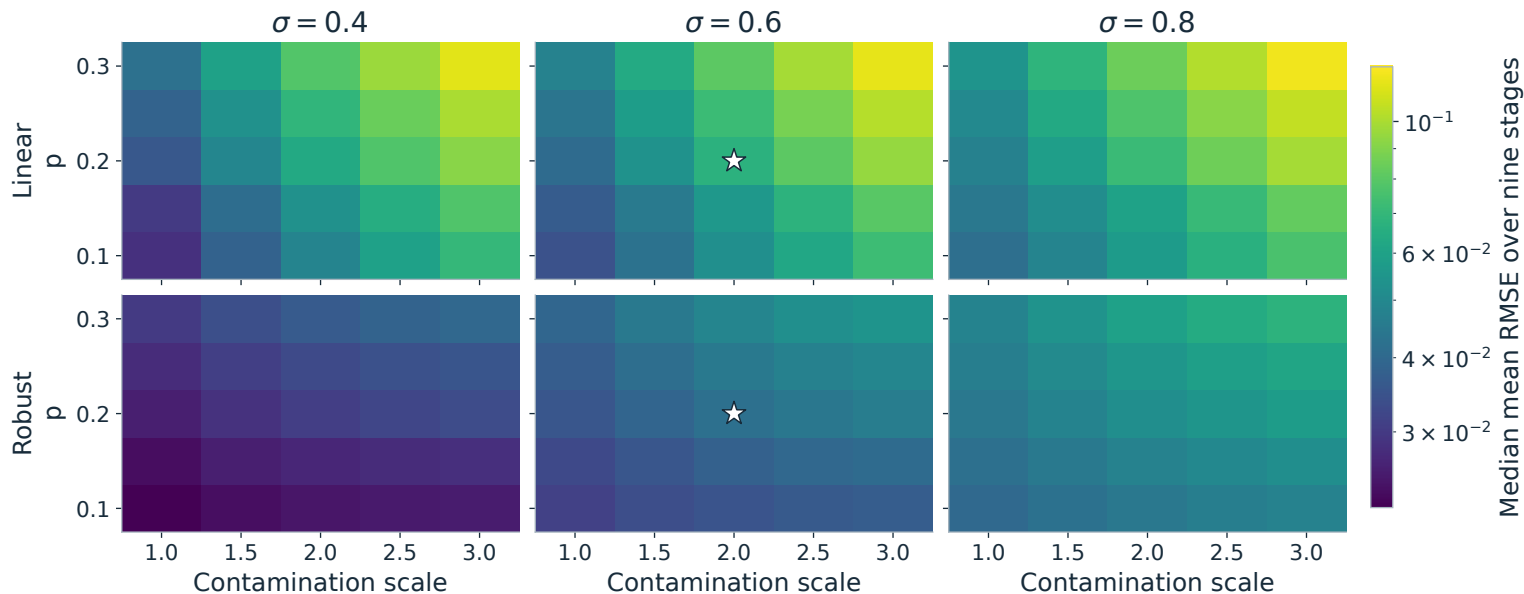
The advantage grows with contamination magnitude



Fixed $\sigma = 0.2$, $p = 0.2$, $H = 0.4$; curves and bands show medians and interquartile ranges across three repeats.

At scale 12σ , robust mean RMSE is **74.6% lower** in the additive model and **77.1% lower** in the masked model.

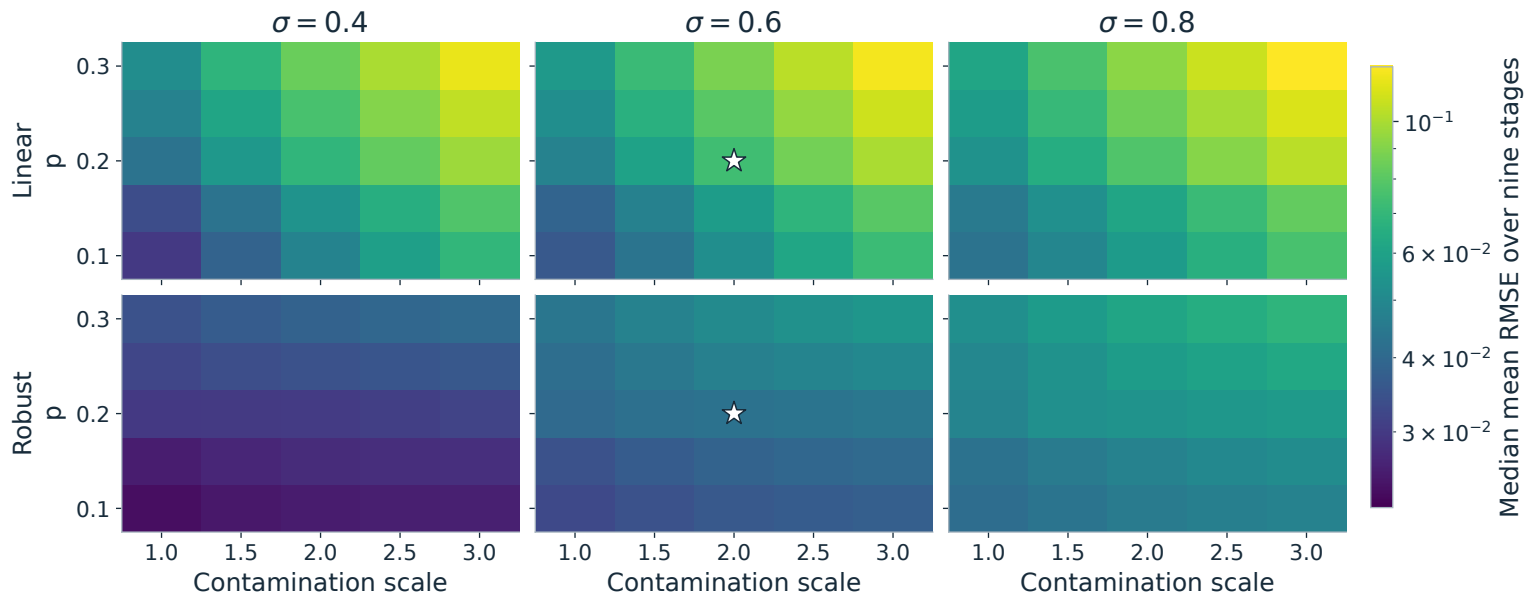
The contamination grid compares both methods



Additive contamination; three repeats per cell. Stars mark the default setting.

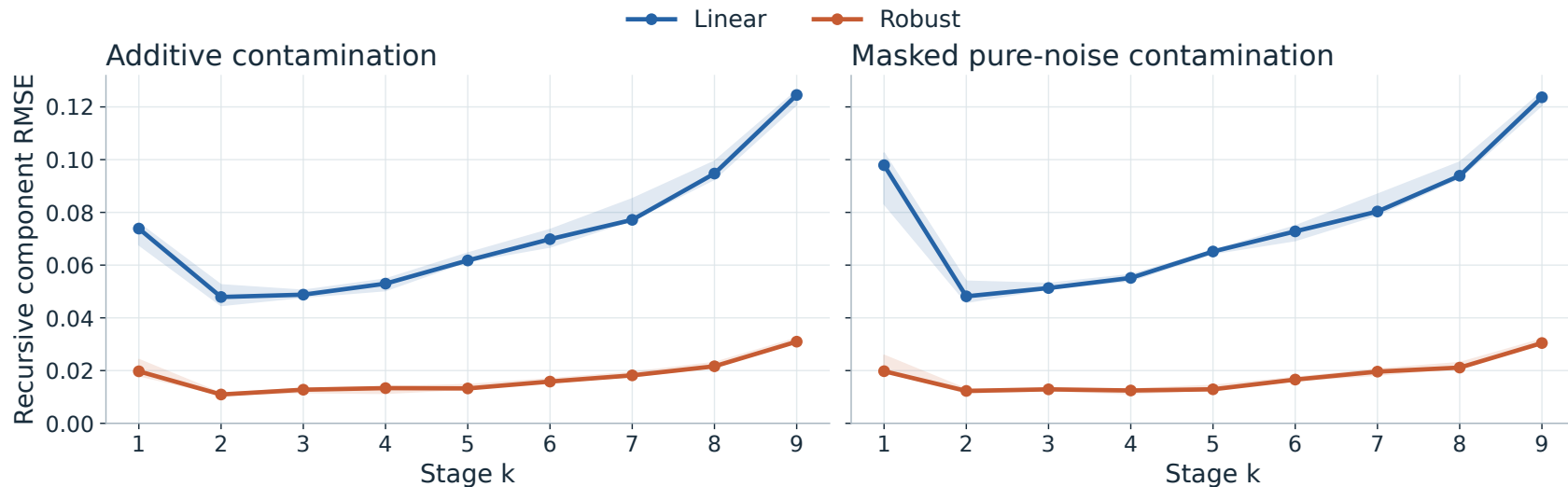
Both heatmaps use one logarithmic color scale for the median mean RMSE over nine stages.

The masked grid compares both methods



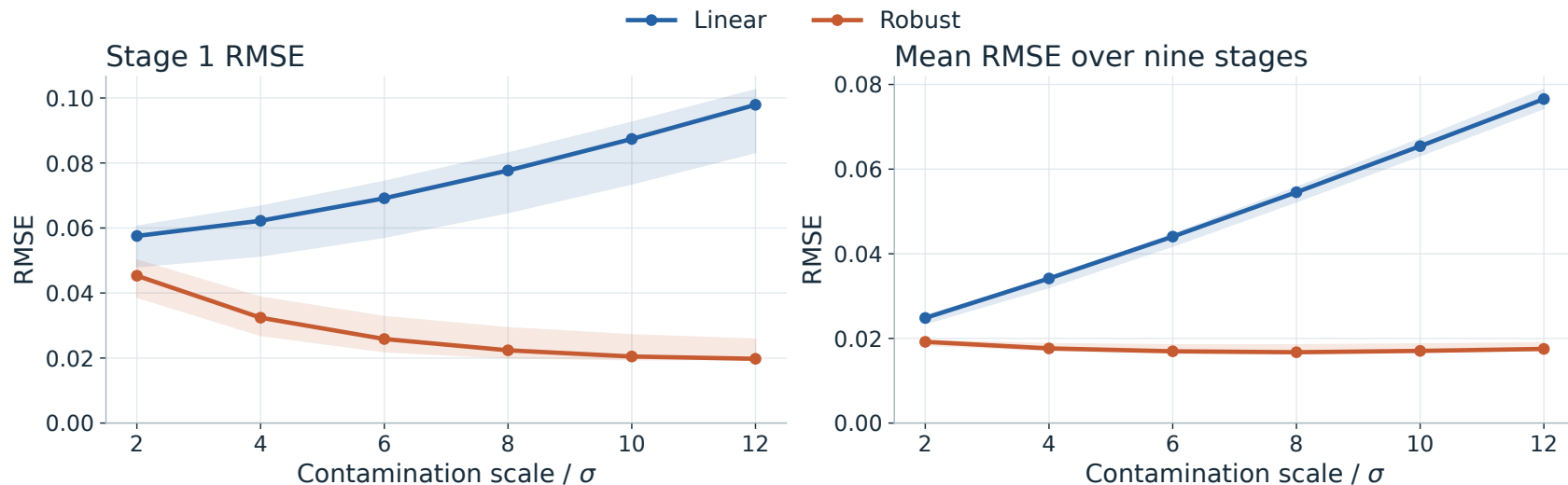
Masked contamination; three repeats per cell. Stars mark the default setting.
Both heatmaps use one logarithmic color scale for the median mean RMSE over nine stages.

Robust gains across all nine stages at 12σ



At $\sigma = 0.2$, $p = 0.2$ and contamination scale $12\sigma = 2.4$, median RMSE over three repeats is lower at all nine stages in both models. Stage-wise reductions range from 73% to 80%.

Larger masked replacements can reduce robust RMSE



Robust mean RMSE falls 13% from 2σ to 8σ , then rises; linear RMSE increases.

Robust stage-1 RMSE falls 56% from 2σ to 12σ , while linear RMSE rises 70%.

Fixed $\sigma = 0.2$, $p = 0.2$, $H = 0.4$; medians and interquartile ranges across three repeats.

Why larger replacements may be less harmful

At a masked point, $Y_i = G_i + C_i$. The clean signal is absent at every contamination magnitude.

Small replacement values

Replacements concentrate near zero. They can pull a local fit toward zero where the clean signal is nonzero.

Large, symmetric replacements

The bounded robust score limits the influence of large residuals. Positive and negative score contributions can cancel.

$$|\psi_H(r)| \leq 1, \quad \frac{\psi_H(r)}{r} \longrightarrow 0 \quad \text{as } |r| \rightarrow \infty.$$

The observed RMSE curve is consistent with this explanation. The experiment does not isolate the mechanism.

The numerical case for robust IMF

- Lower RMSE at all nine stages in both single-run examples.
Mean RMSE is 35.5% lower for additive contamination and 33.2% lower for masked contamination.
- Much larger gains when contamination values are large.
At scale 12σ , mean RMSE falls by about 75% and 77% in the two models.
- The advantage under large contamination spans the full decomposition.
Median RMSE improves at every stage, with reductions of 73% to 80%.

A bounded local score preserves the IMF procedure while reducing sensitivity to large contamination values.

Appendix: implementation details

- Window sizes are 501, 355, 251, 177, 125, 89, 63, 45, 31.
Both methods use periodic wrap boundaries and the same weights.
- The implemented weights are $w(u) \propto (1 - |u|)_+^2$.
The review calls the kernel Epanechnikov; that label differs from the code.
- Robust fits use $H = 2\sigma$, the notebook's lookup grid,
at most 60 gradient-descent iterations and tolerance 10^{-6} .
- RMSE curves and heatmaps use each method's clean reference.
The waveform slide uses one shared clean linear IMF reference.

Main example: $\sigma = 0.6$, $p = 0.2$, contamination scale 2.0; signal seed 2026, observation seed 777. Sweep: $\sigma = 0.2$, $p = 0.2$, scales $2\sigma, 4\sigma, \dots, 12\sigma$, seeds 777000–777002.

Appendix: sources and reproducibility

- IMF.pdf, §2.2 and equation (3.2)
Component definition and recursive update; robust contrast in §3.
- [Source review](#)
Source experiment, contamination models and robust contrast.
- [Additive comparison](#) and [masked comparison](#)
Both methods are recomputed on the same contaminated observation.
- [Additive sweep](#) and [masked sweep](#)
Magnitude slices: 18 trials per model. Heatmaps: 225 trials per model across three noise levels, five probabilities and five magnitudes, with three repeats per setting.

The companion script exports stage RMSE, mean RMSE and vector figures.
CSV data and source hashes are stored beside the figures.

Build instructions: `gd_imf_real_vs_calculated_slides.md`, next to the LaTeX source.